

$$3. H_1: (L \wedge G) \rightarrow H$$

$$H_2: J \rightarrow L$$

$$H_3: Jt \rightarrow J$$

$$H_4: \neg G \wedge G_S$$

$$H_5: Jt$$

$$C: M$$

$$H_1, H_2, H_3, H_4, H_5 \vdash^? C \quad (=)$$

$$(\Rightarrow) \text{CNF}(H_1 \wedge H_2 \wedge H_3 \wedge H_4 \wedge H_5 \wedge \neg C) \vdash^?_{\text{Res}} \square$$

$$H_1 \equiv \neg(L \wedge G) \vee H \stackrel{\text{De Morgan}}{=} \neg L \vee \neg G \vee H$$

$$H_2 \equiv J \vee \neg L$$

$$H_3 \equiv Jt \vee \neg J$$

$$H_4 \equiv \neg G \wedge G_S$$

$$H_5 \equiv Jt$$

$$\neg C \equiv \neg M$$

$$C_1: \neg L \vee \neg G \vee H$$

$$C_2: J \vee \neg L$$

$$C_3: Jt \vee \neg J$$

$$C_4: \neg G$$

$$C_5: \neg G_S$$

$$C_6: Jt$$

$$C_7: \neg M$$

→ Simplify using Davis-Putman transformations.

$$S = \{C_1, \dots, C_7\} \vdash^{\text{res}} \square$$

Eliminate C_5 because G_S is a pure literal in S .

$$\left. \begin{array}{l} \text{Res}_x(f \vee l, g \vee \neg l) = f \vee g \\ \text{Res}_x(l, \neg l) = \square \end{array} \right\} \text{ they resolve upon the literal } 'l'.$$

$$C_8 \equiv \text{Res}_H(C_1, C_7) = \neg L \vee \neg G$$

$$C_9 \equiv \text{Res}_{Jt}(C_3, C_6) = J$$

$$C_{10} \equiv \text{Res}_G(C_8, C_4) = \neg L$$

$$C_{11} \equiv \text{Res}_L(C_2, C_{10}) = \neg J$$

$$C_{12} \equiv \text{Res}_J(C_9, C_{11}) = \square \Rightarrow \text{inconsistency}$$

$$\text{Conclusion: } S \vdash^{\text{res}} \square,$$

$$\text{so } H_1, \dots, H_5 \vdash C.$$

$S = \{p \vee q, \neg p \vee q \vee r, \neg p \vee q \vee r, \neg q \vee r, \neg q \vee r\}$

LOCK RESOLUTION

Prove inconsistency

→ assign indexes in random order. (arbitrary)
 → decrease the no. of resolvents.

(you can try at home a $\neq$ indexing)

The relative order matters, not the value of index!
 (of the literals in the clauses)

Exam: $C_1: p \vee q$
 $C_2: \neg p \vee q \vee r$
 $C_3: \neg p \vee q \vee r$
 $C_4: \neg q \vee r$
 $C_5: \neg q \vee r$

(1) S is inconsistent ($\Rightarrow S \vdash_{\text{lock res.}} \square$)

$p, \neg p$ to have the lowest indexes in their clauses. $\rightarrow C_1, C_2$ or C_1, C_3

$$C_6 = \text{Res}_p^{\text{lock}}(C_1, C_2) = q \vee r \quad (2) (5) \quad (q \text{ inherited from both, but keep the lowest index})$$

$$C_7 = \text{Res}_p^{\text{lock}}(C_1, C_3) = q \vee r \quad (2) (8)$$

$$C_8 = \text{Res}_q^{\text{lock}}(C_6, C_5) = r \vee r \equiv T \quad (12) (5)$$

$$C_9 = \text{Res}_q^{\text{lock}}(C_6, C_4) = r \quad (5)$$

$$C_{10} = \text{Res}_r^{\text{lock}}(C_7, C_5) = r \quad (8)$$

$$C_{11} = \text{Res}_r^{\text{lock}}(C_9, C_{10}) = \square \text{ inconsistent.}$$

(2) Lock resolution + level saturation strategy.

Check the consistency of: $S = \{ \underbrace{q \vee r}_{C_1}, \underbrace{p \vee r}_{C_2}, \underbrace{\neg q \vee r}_{C_3}, \underbrace{\neg p \vee r}_{C_4} \}$

$S^0 = S$ (initial set)

$$S^k = \{ \text{Res}^{\text{lock}}(C_i, C_j) \mid C_i \in S^{k-1}, C_j \in S^0 \cup S^1 \cup \dots \cup S^{k-1} \}$$

if $\square \in S^k$, then S is inconsistent.

if $S^k = \emptyset$, then S is consistent.
 (the last level is empty)

In lock resolution,
 we work with
tautologies!

a) first indexing

$$C_1 = \underset{(1)}{q} \vee \underset{(2)}{r}$$

$$C_2 = \underset{(3)}{p} \vee \underset{(4)}{\neg r}$$

$$C_3 = \underset{(5)}{\neg q} \vee \underset{(6)}{r}$$

$$C_4 = \underset{(7)}{\neg p} \vee \underset{(8)}{r}$$

$S^0 = S$ (initial set)

$$S^1 = \{ \text{Res}^{\text{lock}}(C_i, C_j) \mid C_i \in S^0, C_j \in S^0 \} = \{C_5, C_6\}$$

$$C_5 = \text{Res}_q^{\text{lock}}(C_1, C_3) = \underset{(2)}{r}$$

$$C_6 = \text{Res}_p^{\text{lock}}(C_2, C_4) = \underset{(4)}{\neg r} \vee \underset{(8)}{r} \equiv \top$$

↗ "tautology"

$$S^2 = \{ \text{Res}^{\text{lock}}(C_i, C_j) \mid C_i \in S^1, C_j \in S^1 \}$$

$$C_7 = \text{Res}_r^{\text{lock}}(C_5, C_6) = \underset{(8)}{r} = C_5 \quad (\text{not a new clause!})$$

We cannot resolve C_5 with a clause from S^0 .
 C_6 with a clause from S^0 ?

→ all the lock resolvents were obtained.

$$\Rightarrow S^2 = \emptyset.$$

After a complete search using level-set-strategy, without deriving the empty clause, we conclude that S is consistent.

b) second indexing

$$C_1: \underset{(1)}{q} \vee \underset{(2)}{r}$$

$$C_2: \underset{(3)}{p} \vee \underset{(4)}{\neg r}$$

$$C_3: \underset{(6)}{\neg q} \vee \underset{(5)}{r}$$

$$C_4: \underset{(8)}{\neg p} \vee \underset{(7)}{r}$$

I will not assign the lowest index.

$$\Rightarrow S^1 = \emptyset \Rightarrow \text{consistency.}$$

④ Build the linear refutation from S. (binary tree, graphical repres.)

$$S = \{ \underbrace{\neg p \vee q}_{c_1}, \underbrace{p \vee \neg q \vee r}_{c_2}, \underbrace{\neg r}_{c_3}, \underbrace{p \vee q \vee r}_{c_4}, \underbrace{\neg p \vee \neg q}_{c_5} \}$$

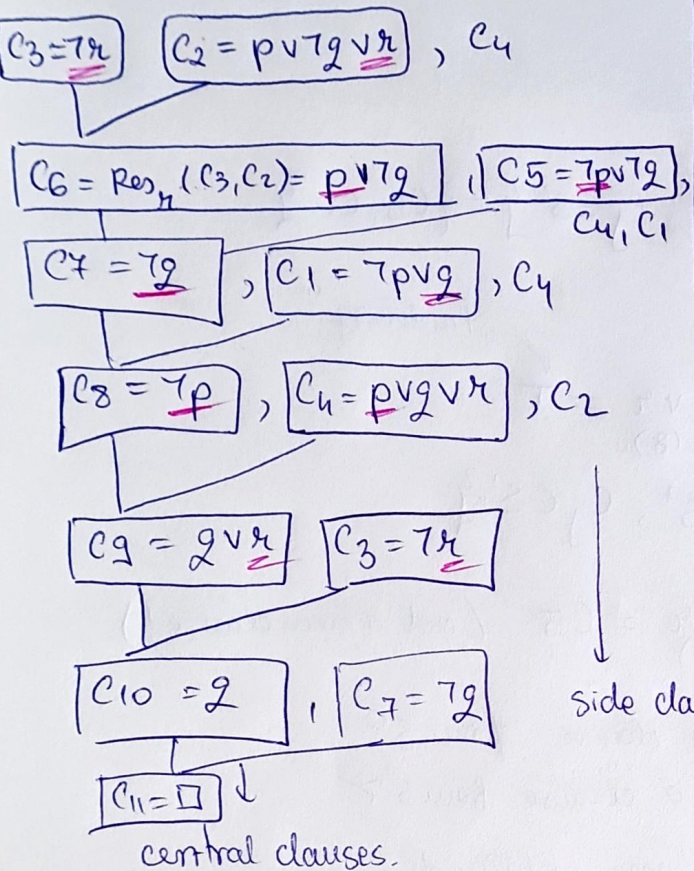
$$c_3 = \neg r \quad \text{TOP CLAUSE}$$

$$c_1 = \underbrace{p \vee q}_{(2) \quad (3)} \quad c_2 = \neg p \vee p \equiv T$$

$$c_3 = \text{Res}_p^{\text{lock}}(c_1, c_2) = \underbrace{q \vee p}_{(3) \quad (6)}$$

Tautologies help us change the relative order of the literals, so don't remove them!

As side clauses, you can use original clauses, along with new clauses, as well.



Conclusion: $S \vdash_{\text{lim. res}} \square$, so S is inconsistent.

(Consistency $\rightarrow$ backtrack.)